InventOR: Can Large Language Models Set Reorder Points and Safety Stocks? Industrial Evidence Against ERP and Data-Driven Benchmarks

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Abstract

When do ERP-relative savings justify operational consideration of training-free large language model (LLM) replenishment proposals? *InventOR* is a training-free screening protocol that tests an unmodified pretrained LLM against comparator dependence, ranking reversals and joint trajectory feasibility on industrial records. Three original generations, seven reference arms and an ERP-input ablation cover 365 plant-material pairs from three plants. Three masked generations reproduce the aggregate rankings on 338 common pairs. On the original 346-pair common-feasible cohort at zero shortage price, LLM policies cost about 11% less than ERP-derived parameters and 30–31% less than code using ERP planning inputs, but 6%, 8% and 18% more than history-only, Croston/SBA and empirical-quantile baselines. Mean fill is 0.1–0.4 percentage points below ERP-derived policies. Plant C and fixed-trajectory shortage repricing reverse the pooled ERP comparison. Static storage checks miss trajectory breaches; capacity clipping releases orders below the minimum order quantity (MOQ). A separate offline intervention reuses frozen policies with strict admission, supported by a conditional invariant and finite-state counterexamples, without guaranteeing service or optimality. A public masked panel, frozen policies and code support numerical replay without fine-tuning or proprietary training data, at an estimated few cents and about

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five minutes per material. ERP-relative savings are therefore necessary but not sufficient: competitive, feasible proposals require the joint screening InventOR provides.

Keywords: Inventory Control, Large Language Models, Generative Artificial Intelligence, Safety Stock, Intermittent Demand, Empirical Evaluation

1. Introduction

Manufacturing replenishment requires plant-material reorder points, safety stocks and order quantities. Revising them requires demand/lead-time estimation, inventory theory, minimum order quantity (MOQ) and storage checks, and planner explanations. LLMs with code interpreters are proposed for supply-chain decisions (Fosso Wamba et al., 2024; Jackson et al., 2024; Song et al., 2026); their numerical proposals still require independent evaluation.

Three problems limit operational consideration of LLM replenishment proposals. First, LLM outputs can be unfaithful or fabricated (Ji et al., 2023), so a proposed policy is a candidate, not a decision. Second, savings over ERP parameters need not survive stronger comparators, plant differences or shortage pricing. Third, static storage compliance need not imply joint MOQ/storage feasibility as stock builds up over time.

The empirical gap is whether ERP-relative savings remain decision-relevant under stronger comparators, plant/price reversals and joint trajectory checks. InventOR evaluates training-free, fixed per-material (r, Q) proposals with explicit input boundaries. Where recent work fine-tunes models, trains on synthetic or proprietary corpora, or co-designs generator/verifier architectures (Section 3), InventOR asks what an unmodified pretrained model can deliver on real ERP data and which checks decide whether its proposals deserve operational consideration. Deterministic software gates and simulates proposals; planners retain authority (Section 2).

We study five questions on real industrial data: what cost, service and inventory the

LLM policies achieve relative to ERP-derived parameters, data-driven baselines and textbook code, with explicit input boundaries (RQ1); how their differences relate descriptively to parameter similarity, run-to-run noise, demand segments and shortage prices (RQ2); whether policies respect storage limits along simulated trajectories (RQ3); how stable they are across repeated generations (RQ4); and which plant and demand-regime results might inform proposed review and escalation rules (RQ5).

We make three contributions. (i) *A training-free screening protocol*: InventOR couples an unmodified pretrained LLM with deterministic gates, seven information-controlled reference arms, an ERP-input ablation and hard-limit trajectory replay, so that LLM replenishment proposals can be judged without fine-tuning, training corpora or solver-specific models. (ii) *Industrial evidence*: on multi-plant production records, ERP-relative savings are real but comparator-, plant- and price-dependent, and joint MOQ/storage failures remain invisible to static checks; savings are necessary, not sufficient. (iii) *A replayable, strict-admission remedy*: frozen original/masked replay and a strict-admission intervention with invariant and counterexample tests (Section 6; Supplementary File S2) remove trajectory breaches while the public masked panel, frozen policies and code allow numerical replay. Unlike fine-tuned optimization models such as ORLM, trained on synthetic instances (Huang et al., 2025), InventOR shows how far an off-the-shelf model already goes on real data and which verification makes its proposals usable.

At zero shortage price original LLM policies cost about 11% less than ERP-derived parameters at mean fill 0.1–0.4 points lower, but 6%, 8% and 18% more than history-only, Croston/SBA and empirical-quantile policies. Code using ERP inputs costs 44–45% more than the LLM. Plant C and shortage repricing of fixed trajectories reverse the ERP comparison. Static-compliant policies exceed storage limits on unconstrained trajectories for

53–66% of pairs; capacity clipping can breach MOQ. Cost advantage and joint feasibility require distinct checks.

2. Problem Setting and Workflow

2.1. The decision problem

The decision supported is the periodic (re)setting of ERP reorder points, safety stocks and order quantities. This partly structured operational-control decision (Gorry & Scott Morton, 1971) combines textbook calculation with judgement about estimators, intermittent demand, recorded lead times and constraints. The material planner decides; minimum order quantity (MOQ) and physical storage constrain proposals. The evaluated scope is single item, single echelon, daily working-observation review, lost sales and the (r, Q) family. Live order release, supplier selection, forecasting-model development and multi-echelon networks are out of scope. The LLM proposes candidates; it does not place orders, decide deployment or run the deterministic held-out evaluation.

2.2. The InventOR workflow and its level of automation

> **Definition.** *InventOR is a proposed human-reviewed planning workflow for each plant-material pair: (i) a prompt-configured LLM proposes an (r, Q) policy, justification and escalation flag; (ii) deterministic software gates it against explicit constraints; (iii) offline simulation compares it with reference policies on held-out recorded series; and (iv) a planner would review the result and alone decide ERP changes.*

The study executes generation, deterministic gating and offline evaluation; planners at the case company reviewed the proposals (Section 7). In the levels of automation of Parasuraman et al. (2000), the proposed workflow would operate at a low-to-intermediate level, leaving

execution to humans. Section 4.3 proposes safeguards against complacency and automation bias (Parasuraman & Manzey, 2010).

2.3. Requirements on the workflow

The requirements follow from the known unreliability of LLM outputs (Ji et al., 2023), guidance on human–AI interaction and accountability (Amershi et al., 2019; Raji et al., 2020), run-to-run nondeterminism of LLM inference (Ouyang et al., 2025) and the constraints of inventory control. Table 1 ties each to a design feature (Section 4) and an evaluation (Section 6).

Table 1: Requirements on the workflow, the features that address them and where they are evaluated.

ID	Requirement	Design feature	Evaluation
DR1	Bounded generation: the LLM proposes and has no authority	Narrow candidate-generator role; prohibited methods; no order placement	Sections 6.1, 6.3
DR2	Verifiable feasibility along time, not only at a point	Deterministic gate; hard-limit trajectory replay	Sections 6.1, 6.5
DR3	Counterfactual evidence for the reviewer	Held-out simulation against reference arms	Sections 6.2, 6.10
DR4	Information transparency: comparisons control what each arm sees	Seven reference arms; LLM ablation without ERP parameters	Sections 6.3, 6.4
DR5	Reproducibility and accountability of the configuration	Frozen prompt file; per-call log; repeated runs	Section 6.8
DR6	Human control with explicit escalation	Trust, escalate and reject rules; escalation flag	Sections 4.3, 6.6

3. Related Work and Positioning

Industrial inventory evidence and validation.. Bai et al. (2012) study a practical batch-ordering policy under incomplete demand knowledge; Fu and Chien (2019) link intermittent-demand forecasting to an electronics-distribution application. These motivate implementable comparators and explicit information boundaries, not the assumption that better forecasts ensure better policies. Lau et al. (2008) compare inventory policies under simulated operating conditions, supporting conditional cost/service interpretation. InventOR likewise

distinguishes estimation, static feasibility and realized replay outcomes; its recorded-series, lost-sales replay is not a replication of those experiments.

Human–AI decision making in planning.. Decision support aims to improve human judgement, not replace it (Gorry & Scott Morton, 1971; Shim et al., 2002; Sprague, 1980); suggestion scope and verification influence appropriate reliance (Parasuraman & Manzey, 2010; Parasuraman et al., 2000). Using a retail-company case and simulation, X. Wang et al. (2024) document algorithm aversion in safety-stock decisions during disruptions and report better algorithmic performance in that setting. People also discount algorithms after seeing them err (Dietvorst et al., 2015). These findings do not establish benefits from InventOR, which retains planner authority.

LLMs in operations and supply-chain decisions.. Wu (2026) compares automated and human-specified LLM-assisted inventory optimization under disruptions. Zabraoui et al. (2026) address conversational inventory analytics with an EOQ module. Y. Wang et al. (2025) separate candidate generation from scheduling evaluation; El Baz et al. (2026) combine agentic warehouse optimization and simulation. Yan and Ji (2026) describe mechanism-aware multi-agent scheduling with generation–verification, and SimGPT integrates Simio with LLMs across simulation generation, execution and analysis (Dehghani, 2026). InvEvolve's arXiv v4 preprint describes reinforcement-learning-trained LLMs and online inventory-policy evolution with confidence-based certification (Huang et al., 2026). These are preview/abstract-supported descriptions, not independent verification of their methods or guarantees. InventOR instead evaluates fixed prompt-configured proposals without project-specific training. Reviews map generative AI opportunities in supply chains (L. Li et al., 2024; Song et al., 2026) and emphasize human roles, data quality and governance (Cannas et al., 2024; Trifone et al., 2026; Venkatesh et al., 2024). Examples

include fine-tuned digital-manufacturing optimization (Amarasinghe et al., 2026) and learned order timing and quantity (Liu et al., 2026). Optimization modeling/code generation (Ahmaditeshnizi et al., 2024; Huang et al., 2025; B. Zhang et al., 2025) differs from OptiGuide's natural-language explanation and what-if interaction with solver outcomes (B. Li et al., 2023), and from multi-agent inventory control (Jannelli et al., 2025; Quan & Liu, 2024). InventoryBench evaluates OR algorithms, LLM agents and humans on more than 1,000 instances, reporting benefits from OR-augmented LLMs (Baek et al., 2026). Our narrower question concerns fixed policies on an industrial extract; it requires no training corpus, fine-tuning or solver-specific model and can be replayed from public masked inputs. In a controlled retail-forecasting experiment with 123 human forecasters and five LLMs, Abolghasemi et al. (2025) find that the tested LLMs do not consistently outperform humans. Abbaszadeh Nakhost et al. (2025) examine LLM-based spare-parts demand forecasting. We test policy variability directly (Section 6.8), rather than infer it from forecasting results.

Data-driven inventory control without LLMs.. Learning-based replenishment includes non-stationary demand policies (Ren et al., 2024), deep reinforcement learning for multi-echelon systems (Kolyaei et al., 2025; Stranieri et al., 2024), contextual robust optimization (Q. Wang et al., 2026) and forecasting-linked reinforcement learning (Schett et al., 2026). Gabellini et al. (2026) evaluate joint replenishment, safety-stock and safety-lead-time decisions on real automotive-company data. Forecasting research links forecasts to inventory costs (Kourentzes et al., 2020), and TFT-MPIR generates multi-period orders from historical/contextual inputs (Guo et al., 2025). InventOR evaluates a fixed-prompt pretrained generator that needs no training.

Intermittent and spare-parts demand.. Manufacturing histories are often intermittent or lumpy. Work began with the bias of Croston's estimator and its stock-control performance

(Syntetos & Boylan, 2001, 2006) and continues with neural (Kourentzes, 2013) and state-space (Svetunkov & Boylan, 2023) forecasts and order-up-to control (Rostami-Tabar & Disney, 2023), all of which bear on the inputs of the policies we compare. The wider literature adds specialized estimators and classification schemes (Babai et al., 2026; Croston, 1972; Prak et al., 2021; Syntetos & Boylan, 2005). InventOR's (r, Q) policy is the standard workhorse of continuous-review control (Hadley & Whitin, 1963; Porteus, 2002; Silver et al., 2017).

4. The InventOR Workflow

4.1. Architecture and components

Following classical decision-support components (Shim et al., 2002; Sprague, 1980), InventOR comprises *data* (material-scoped histories, costs, constraints and ERP values; DR4); *knowledge* (the frozen task, prohibited methods, rules and schema in Supplementary File S1; DR1, DR5); *models* (a prompt-configured enterprise LLM/code-interpreter generator and a separate deterministic parser, gate and held-out simulator; DR1–DR3); *dialog* (policy, justification, flag and comparison for proposed planner review; DR6); and *governance* (input hashes, timestamps, sessions, raw responses and repeats; DR5). The generator is non-inspectable. The evaluated dialog is a per-pair output record; no interactive interface was built or evaluated.

4.2. Operation in four decision steps

For one material, InventOR uses four decision steps after input preparation; Figure 1 includes preparation as a fifth box.

Step 1. Generate. The enterprise LLM/code interpreter receives one material's history and cost, constraint, service-target and ERP parameters (Section 5.2). The written

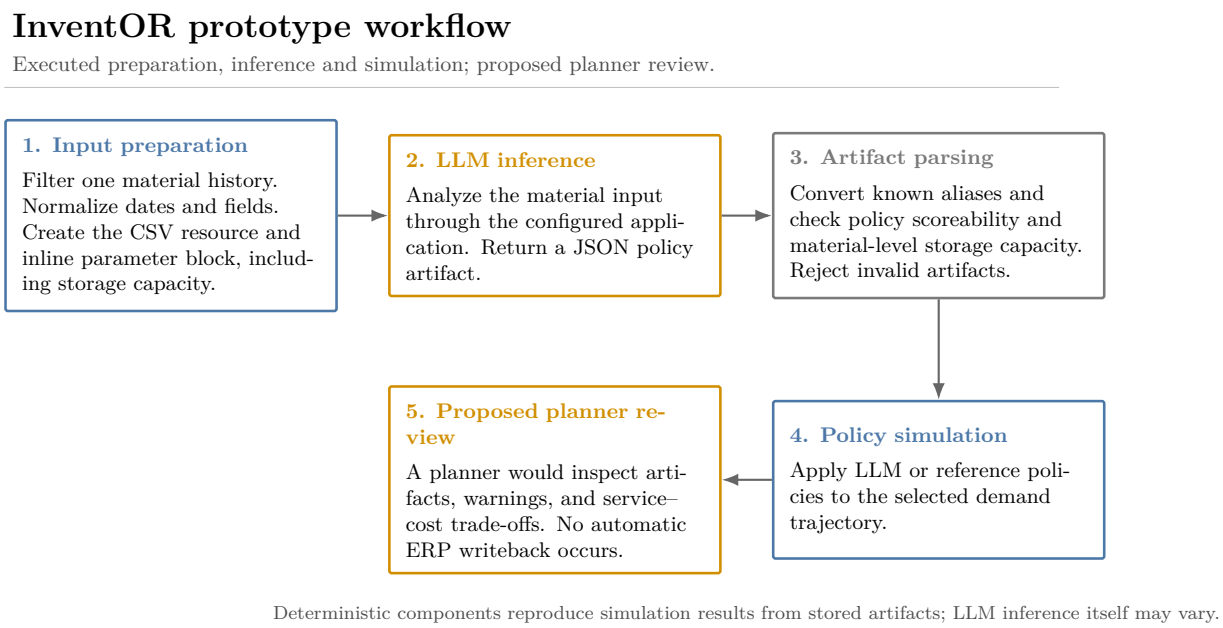


Figure 1: InventOR workflow: material data are prepared, an LLM proposes a policy, and deterministic software parses, checks and simulates it. Planner review is a proposed deployment step, not an evaluated activity.

prompt is the only configuration; weights remain unchanged (Sahoo et al., 2024). It requests transparent statistics, a structured policy, justification and escalation flag, and gives an EOQ-related quantity rule with MOQ/storage bounds. It does not prescribe the normal-approximation safety-stock formula used by comparators. All scored artifacts resolve to fixed (r, Q) policies; permitted family labels are not separate evaluated methods.

Step 2. Gate. Deterministic code extracts the policy, rejects malformed or infeasible outputs (missing or non-numeric fields, non-positive reorder point or order quantity, negative safety stock, reorder point below safety stock, safety stock plus order quantity above the storage limit) and retries the generation up to three times. The backtest applies one further admission check, that the order quantity is not below the MOQ.

Step 3. Simulate. Each accepted policy is replayed on the material's demand observed

after the data cutoff, a held-out evaluation in the sense of out-of-sample forecast testing (Hewamalage et al., 2023; Tashman, 2000), together with the comparator policies, producing holding and ordering cost, fill rate and stockout days.

Step 4. Review (proposed deployment step). A planner would inspect the policy, justification, escalation flag and simulated comparison and would retain sole authority over ERP changes. Recording the planner's decision, review time and comments is proposed future functionality; neither this logging nor planner outcomes was evaluated in the present study.

The LLM is a *candidate generator* whose output is gated; it does not run the study's deterministic held-out evaluation, place orders or decide ERP changes. The prompt prohibits Monte Carlo simulation, distribution fitting, machine-learning libraries and specialized intermittent-demand estimators, so that the policy rests on transparent statistics.

4.3. Trust, review and reject rules

We propose three rules for planners, in line with guidelines for human–AI interaction and AI auditing (Amershi et al., 2019; Raji et al., 2020). The rules and the 0.5-point threshold are design proposals not evaluated with planners; the trust rule would be met by a minority of pairs (the LLM is cheaper than the history-only OR in 74–77 of 346).

Trust (consider after review) only after checking both order-level MOQ and storage compliance, as well as a simulated cost no higher than the best history-only reference (Croston/SBA or empirical quantile) with fill not more than about 0.5 points lower. Capacity-clipped replay alone cannot establish the joint compliance needed to apply this rule.

Escalate when the LLM raises an escalation flag, the material is lumpy or has strongly seasonal or trending demand, the plant or calendar data were repaired, or the policy differs

strongly between repeated generations.

Reject when the proposal fails the gate after the retry budget, exceeds the storage limit in the hard-limit replay, or is dominated (higher cost and lower fill) by a history-only reference.

4.4. Provenance and reproducibility of the configuration

The system prompt, deployed unchanged, defines the task, forbidden methods, capacity rules and output schema (Supplementary File S1; excerpts in Appendix B). Calls log input SHA-256, UTC time, session identifier and raw response. The platform does not expose model identifier or sampling settings; Anthropic Claude Sonnet 4.6 with a code interpreter is operator-declared. Frozen policies support numerical replay, not regeneration of proprietary inference. Three repeated generations examine within-panel variation (Section 6.8).

5. Application and Evaluation Design

5.1. Evaluation approach

The evaluation addresses Table 1 through hard-limit replay with working-day receipts and no shortage penalty, plus sensitivities. This ex post simulation replays the processed operational-extract series, which the data owner confirms records complete actual demand (consumption) per material. There are no live users. The history-only, SLT-informed and ERP-floor OR arms implement textbook EOQ and normal-approximation policies with different ERP-input treatments, not exact execution of a prescribed prompt formula. Outcomes characterize conditional policy cost, service and constraint failures, not usability or planner benefits.

5.2. Industrial application and data

The data come from a large manufacturer, and the evaluation is retrospective. The dataset contains 233,078 daily records for 411 plant-material pairs from three plants (A,

B and C), with a data cutoff on 10 March 2026 and a post-cutoff horizon observed to 27 August 2026. Eligibility requires at least 20 working-day observations before the cutoff and 40 on or after it within the evaluation horizon. Of the 411 pairs, 365 are eligible for scoring (234 in Plant A, 65 in Plant B, 66 in Plant C) and are sent to the LLM. For 19 pairs (all in Plant A), at least one of the SAP-derived or SLT-informed reference policies cannot be projected to satisfy MOQ and the static $SS + Q$ capacity rule while retaining its safety stock; these are excluded from outcome comparison, leaving a *common-feasible cohort* of 346 pairs (215, 65 and 66). This is a comparator-specific exclusion, not proof of general policy infeasibility. Demand is concentrated (the five largest materials account for 40.3% of demand). The source working-day flag of Plant C is all zero; we reconstruct it from the same-date majority of Plants A and B and test this repair in Section 6.10. Data quality is a known threat to decision support (Watts et al., 2009) and inventory records (DeHoratius & Raman, 2008; Kang & Gershwin, 2005; Mersereau, 2013).

The extract has 220,017 distinct plant-material/date groups. A masked-panel diagnostic finds 12,036 repeated groups across 62 of the full panel's 411 pairs: daily quantities repeat unchanged while context varies. Frozen scoring sums signed same-date demand before taking its absolute value, averages positive lead times and retains the last row's other attributes; stock is not summed. These are documented, reproducible processing choices. The implementation is in `inventory_llm/core.py`: `load_grouped_rows`, `demand`, `compute_stats` and `eligible_item_keys`.

Each pair is sent as a material-scoped CSV with daily demand, recorded lead times, a working-day flag, the minimum order quantity, unit cost, holding-cost rate, ordering cost, service-level target, the storage limit, and two *ERP planning parameters* (current safety stock and safety lead time), which are withheld in the ablation arm. The CSV also carries

supplier and lane context fields that the prompt does not use. A masked version of the panel is released for replication (Section 6.9); there the plants are labelled PL1, PL2 and PL3 (corresponding to A, B and C).

5.3. Reference arms and information content

ERP safety stock and safety lead time embed planner judgement and are alternative ways of buffering unreliability (van Kampen et al., 2010), and planned lead times can drift upward (Thürer et al., 2023). The LLM and the classical comparators therefore do not necessarily use the same information, and we construct arms that make it explicit (Table 2).

Table 2: Comparison arms and their information. Policy quantities satisfy MOQ and are projected to the storage limit when possible; capacity-clipped orders in replay can fall below MOQ.

Arm	Parameters from	ERP inputs used	Purpose
SAP-derived	ERP safety stock and safety lead time; EOQ order quantity	Both	ERP-derived reference (not observed practice)
SLT-informed OR	Stationary (r, Q) with ERP safety lead time added to the lead time	Safety lead time	OR policy using the ERP safety lead time
ERP-floor OR	SLT-informed OR with safety stock floored at the ERP value	Both	OR arm matched to the LLM's full information
History-only OR	Stationary (r, Q) from demand and lead-time history, mean lead time	None	ERP-independent baseline
Croston/SBA OR	(r, Q) with the Syntetos–Boylan-approximated Croston demand rate and its forecast-error spread, mean lead time	None	Intermittent-demand baseline
Empirical-quantile OR	(r, Q) with the reorder point set to the service-level quantile of historical rolling lead-time demand	None	Distribution-free baseline
Simulation-tuned (r, Q)	Reserve factor and order-quantity multiple chosen by simulation on the last 250 working days before the cutoff	None	Trained data-driven baseline
LLM (runs 1–3)	LLM proposal with all CSV fields (Section 5.2)	Both	Main treatment
LLM without ERP	LLM with ERP safety stock and safety lead time set to "not supplied"	None	Ablation: what the LLM does when it cannot anchor on ERP

Three further arms use no ERP planning inputs. *Croston/SBA OR* uses bias-corrected

Croston demand (Croston, 1972; Syntetos & Boylan, 2005), smoothing 0.1, and root mean squared one-step forecast error as demand spread, a study-specific heuristic rather than a validated lead-time-demand distribution. *Empirical-quantile OR* uses the service-target quantile of rolling demand over rounded mean lead time. *Simulation-tuned* (r, Q) searches reserve factors $-0.5, -0.25, \ldots, 3.5$ and quantity multiples $0.5, 0.75, 1, 1.5, 2$ (up to 85 candidates), selecting the cheapest meeting the service target, or the best fill if none meets it. Statistics use all pre-cutoff data; tuning replays the last 250 working days without capacity clipping, initialized at each candidate's $R + Q$ with an empty pipeline. This is in-sample tuning, not nested validation; evaluation instead uses shared observed opening stock and capacity clipping. No post-cutoff demand informs selection. Deep reinforcement learning (Kolyaei et al., 2025; Stranieri et al., 2024) is not benchmarked.

5.4. Simulation, capacity replay and metrics

Each pair is replayed as a single-item, single-echelon (r, Q) system (Axsäter, 2015; Hadley & Whitin, 1963; Porteus, 2002; Silver et al., 2017; Zipkin, 2000) with lost sales (Bijvank & Vis, 2011), discrete working-day review, shared observed opening stock, an empty pipeline and deterministic mean-lead-time arrivals (Appendix A). This is simulation, not reconstruction of actual orders (Grasso & Taylor, 1984). Fill and shortage cost condition on the recorded demand series; historical lost sales are not reconstructed. Policy quantities satisfy MOQ (Heese, 2025), but capacity-clipped orders (X. Zhang et al., 2024) may not.

The fixed finite-panel estimand conditions on recorded inputs and frozen policies. Costs use pair-specific horizons without exposure normalization; cost/fill means give equal weight per pair, not per plant. Volume-weighted fill is total served divided by total recorded units, undefined at zero total. Stockout days are summed. Contrasts use ratios of mean costs, not means of pairwise percentages. Intervals for paired mean cost/fill differences use a 5,000-

resample percentile-bootstrap of pairs (95%; seed 20261006): descriptive pair-composition sensitivity, not model-generation uncertainty or population inference; within-plant dependence is not modeled.

A static check $SS + Q \leq$ `max_storage_units` does not show that storage limits hold as inventory evolves, because the reorder point includes lead-time demand and inventory on order can exceed the limit on arrival. We therefore record the post-receipt on-hand stock for every policy and day and report the share of pairs and days above the limit (the *unconstrained* simulation), and we *replay* every policy with each order truncated to the storage limit minus the current inventory position (the *hard-limit replay*), which is our primary evaluation.

5.5. Repeated runs and sensitivities

To examine run-to-run variation, the entire cohort is generated three times (runs 1, 2 and 3) with the same prompt file and the same data. Run 1 serves as the reference for tabulated results unless stated otherwise; runs 2 and 3 provide replicates. Sensitivity analyses vary the working-day reconstruction (majority, any-plant, strict-all, exclusion of Plant C and calendar-day arrivals) and the shortage penalty (zero, one and five times unit cost).

6. Results

6.1. Completion and policy characteristics

All four original-data runs (three full-input and one ablation, Section 6.4) return an (r, Q) policy satisfying the numerical and static storage gate for all 365 pairs after at most three attempts per pair. The validity gate rejects non-positive reorder points and order quantities, negative safety stocks and reorder points below the safety stock. The ERP safety stock is zero for 198 of the 365 pairs (188 of the 346 in the cohort below), so for a majority of pairs the SAP-derived arm holds no safety stock at all.

6.2. Headline comparison with ERP-derived parameters

Unless stated otherwise, hard-limit replay uses 346 common-feasible pairs, working-day receipts and zero shortage price (unconstrained: Section 6.5). Table 3 compares LLM cost using the reference-arm mean as denominator (positive: LLM costs more). Run 1 costs 11.4% below SAP-derived and 30.5% below SLT-informed OR (means 2,928 versus 3,305 and 4,211); mean fill is 0.18 points below SAP-derived (98.11% versus 98.30%; interval −0.54 to +0.10). Paired mean cost difference is −377 (95% interval −651 to −154). LLM is cheaper in 241 pairs, simultaneously no costlier and no worse in fill in 226. Stockout days increase (910 versus 804). Volume-weighted fill is 78.7–79.3% for all arms except simulation-tuned (84.8%; Table 3).

Relative to SAP-derived parameters the LLM carries about 17% less average inventory (16,256 versus 19,565 units) and about 16% lower holding cost at about the same mean number of orders per material over its evaluation horizon (17.3 versus 17.6), and it accepts 11–20% more stockout days (runs 3 and 2). The saving is therefore a trade of inventory for service, not a better ordering rhythm, and Section 6.10 shows that once shortages carry a cost this trade no longer favours the LLM.

6.3. Information asymmetry: what the LLM adds beyond the ERP inputs

First, the SLT-informed OR and its ERP-floor variant are the most expensive arms (4,211 and 4,248): the ERP safety lead time inflates their effective lead time and buffers. Against them the LLM is 30–31% cheaper in every run, which mostly measures how heavily ERP inputs over-buffer.

Second, the *history-only OR*, which uses no ERP inputs, costs 2,764 at a fill of 98.14%. The LLM costs 5.9–6.0% more in all three runs (run 1: +164, interval +91 to +240), with

Table 3: Hard-limit replay, common-feasible cohort ($n = 346$), working-day receipts, zero shortage penalty. Cost is mean material-horizon holding-plus-ordering cost; inventory is in units. Orders are mean counts per material over its evaluation horizon; stockout days are summed. The last column gives LLM cost relative to the arm in runs 1/2/3 (positive: LLM costs more).

Arm	Mean cost	Holding	Ordering	Avg. inv.	Orders	Mean fill (%)	Aggr. fill (%)	Stockout days	LLM vs arm (%)
ERP-derived (SAP)	3,305	2,172	1,132	19,565	17.6	98.30	79.34	804	−11.4 / −11.3 / −11.4
SLT-informed OR	4,211	2,608	1,603	21,614	24.5	98.23	79.15	831	−30.5 / −30.4 / −30.5
ERP-floor OR	4,248	2,633	1,616	21,937	24.7	98.23	79.15	831	−31.1 / −31.0 / −31.1
History-only OR	2,764	1,725	1,039	15,972	16.3	98.14	78.72	874	+5.9 / +6.0 / +5.9
Croston/SBA OR	2,705	1,665	1,040	15,521	16.3	98.30	78.87	803	+8.2 / +8.4 / +8.3
Empirical-quantile OR	2,489	1,543	946	13,331	14.8	98.35	78.87	747	+17.7 / +17.8 / +17.7
Simulation-tuned (r, Q)	3,266	2,473	793	11,914	12.6	96.05	84.84	1,973	−10.4 / −10.3 / −10.3‡
LLM (run 1)	2,928	1,824	1,104	16,256	17.3	98.11	78.87	910	
LLM (run 2)	2,931	1,850	1,081	16,599	17.0	97.89	78.70	968	
LLM (run 3)	2,929	1,841	1,088	16,958	17.0	98.16	78.90	890	
LLM without ERP	3,014	1,902	1,112	16,507	17.3	98.06	78.89	890	−8.8† (vs SAP)

† Other comparisons of the ablation are in Section 6.4. ‡ Different operating point (mean fill 96.05%, aggregate fill 84.8%); the paired interval of the cost difference includes zero, and with priced shortages this policy is the cheapest arm (Section 6.10).

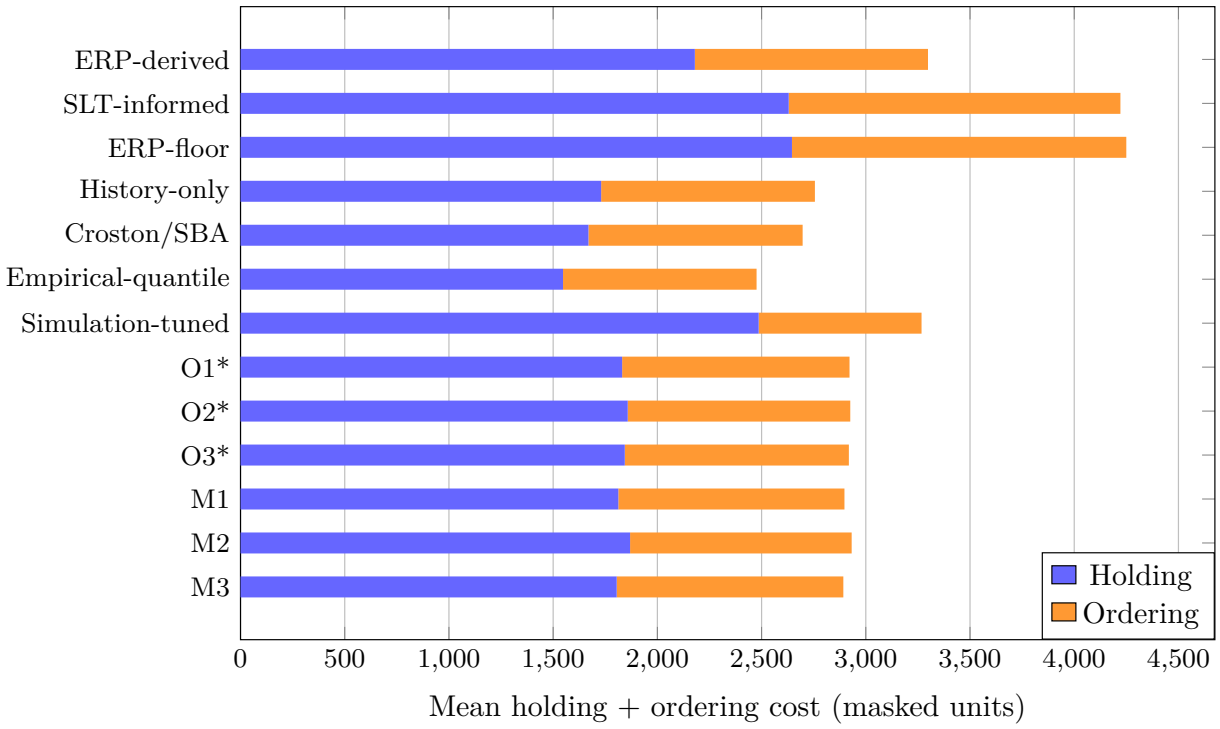


Figure 2: Six-run cost decomposition on the same 338 feasible masked-panel pairs (zero shortage penalty). O1*–O3* are rescaled frozen original policies, not fresh inference; M1–M3 are new masked generations. Comparators are identical across replays. Reciprocal quantity/price scaling preserves economic products before rounding; these costs do not demonstrate monetary confidentiality. Table 3 uses the original cohort/protocol.

fill differences of −0.03, −0.25 and +0.01 points whose intervals include zero, and is cheaper in only 74–77 of 346 pairs. The stronger data-driven arms widen the gap: the LLM costs 8.2–8.4% more than the Croston/SBA OR (2,705, fill 98.30%; run 1: +223, interval +97 to +362; cheaper in 91–99 pairs) and 17.7–17.8% more than the empirical-quantile OR (2,489, fill 98.35%; +439, interval +329 to +557; cheaper in 44–49 pairs; mean fill 0.24 points lower). Part of the LLM's advantage over ERP-derived parameters therefore reflects weak ERP benchmarks.

The simulation-tuned (r, Q) policy sits at a different operating point: an aggregate fill of 84.8% and the lowest inventory (11,914 units) but a mean fill of 96.05% and 1,973 stockout days, because it is tuned on the last 250 pre-cutoff working days. The LLM costs 10.3–10.4% less at zero penalty with a 2.1-point higher mean fill, but the interval includes zero (−2,527 to +857); once shortages are priced the tuned policy is by far the cheapest arm (Section 6.10).

6.4. LLM without ERP planning parameters (ablation)

This ablation blanks ERP safety stock and safety lead time, marking them "not supplied", with the same prompt and other inputs. All 365 original pairs returned a static-compliant policy. Median parameter ratios to run 1 are 1.00, but reorder points differ by more than 10% in 54.6% of pairs, against 48.3% and 48.0% between full-input runs. Hard-limit cost is 3,014 at mean fill 98.06%: 8.8% below SAP-derived (interval −566 to −61), but 9.0% above History-only (interval +134 to +387), 11.4% above Croston/SBA and 21.1% above empirical-quantile. Relative to SAP-derived, costs are −10.8%, −21.9% and +41.6% in Plants A/B/C. The Plant C reversal therefore persists without these fields; one stochastic ablation does not isolate their causal effect or prove that the LLM ignores them. Costs, MOQ, service target and capacity remain available to all arms.

6.5. Storage limits along simulated trajectories

All 365 original LLM policies satisfy $SS + Q \leq$ `max_storage_units`, yet unconstrained post-receipt stock exceeds the limit for 183 pairs (52.9%; 178–183 across runs; 10.2% of pair-days) for the LLM, 198 (57.2%) for SAP-derived and 230 (66.5%) for SLT-informed OR. The static rule does not bound pipeline accumulation and is insufficient for trajectory feasibility. In the hard-limit replay, clipping keeps every arm within one unit of the limit (integer truncation of inventory position), but clipped orders may fall below the MOQ. Enforcing capacity raises LLM mean cost from 2,850 to 2,928 and SAP-derived from 3,184 to 3,305 without changing the ranking.

6.6. Plants and demand regimes

The average hides a strong plant effect. In original run 1 the LLM costs 14.1% less than SAP-derived in Plant A (cheaper in 179 of 215 pairs) and 22.7% less in Plant B (54 of 65), but 36.2% *more* in Plant C (cheaper in 8 of 66); the Plant C loss is similar in the other full runs (+32.2%, +31.5%) and persists under every working-day reconstruction in which Plant C can be scored (Section 6.10). Figure 3 separately shows all six runs on the matched masked-scale cohort. Against the data-driven baselines the original run 1 LLM costs more in every plant: 7.2%, 3.6% and 6.7% more than the history-only OR, 13.1%, 2.3% and 6.2% more than the Croston/SBA OR, and 20.8%, 7.7% and 29.5% more than the empirical-quantile OR in Plants A, B and C. The SLT-informed and history-only ORs coincide in Plant C because ERP safety lead time is zero for all its pairs.

By demand regime, a study-specific high-variability proxy (pre-cutoff working-day demand coefficient of variation CV ≥ 1) identifies 33 materials in the 346-pair original cohort. Labelled "lumpy" in the evidence code, this is not the conventional ADI/CV^2 taxonomy. They show the largest reduction against SAP-derived (-25.4% in run 1), but the history-

only and Croston/SBA ORs cost about 9–10% less than the LLM there; with 33 materials this is indicative only.

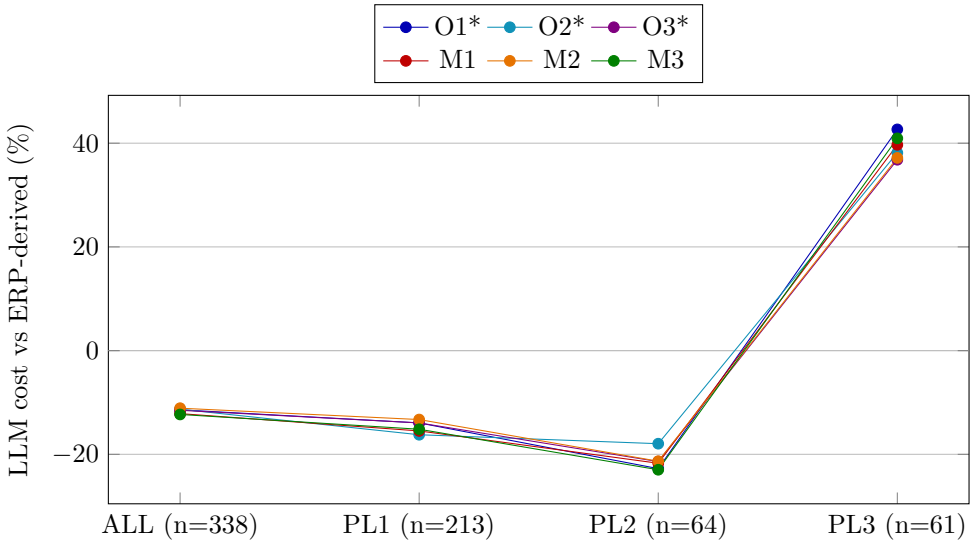


Figure 3: Plant heterogeneity across all six runs on a shared masked-panel cohort. Positive values mean the LLM costs more than ERP-derived parameters. O1*–O3* denote rescaled frozen original policies; M1–M3 denote fresh masked generations. PL1/PL2/PL3 correspond to Plants A/B/C. Lines connect descriptive group summaries, not time points or uncertainty intervals.

6.7. Descriptive comparison with formula-based code

The deterministic History-only OR uses textbook EOQ with an MOQ floor and normal-approximation safety stock (Appendix A); the SLT-informed and ERP-floor ORs additionally use ERP inputs available to the LLM (Section 5). The historical prompt constrains the economic quantity and permits several policy families; it does not prescribe these comparator formulas in full. This is a descriptive comparison with reference code, not a test of compliance with a prescribed normal-approximation recipe.

The LLM resembles the formula-based policies.. The median smoothed ratio $(x_{\text{LLM}} + 1)/(x_{\text{code}} + 1)$ is 1.00 for order quantity (medians 2,810 and 2,828 units) and safety stock (712 and 718, against an ERP median of 0); LLM parameters are averaged over the three original runs first. Here "code" denotes History-only OR. The corresponding reorder-point

ratio is 1.09, including across the segments in Table 4, where LLM cost is 4–9% higher. These summaries indicate similarity and a reorder-point uplift, not item-level proof that the LLM executes the formulas faithfully.

Input treatment changes the ranking against code.. Code that receives the same ERP inputs as the LLM costs 4,211 (SLT-informed OR) and 4,248 (ERP-floor OR) per pair against 2,928 for the LLM in run 1, i.e. 44–45% more using the LLM mean as denominator, because ERP safety lead times inflate the textbook reorder point. Code ignoring ERP planning inputs costs 2,764; the LLM costs 5.9% more using that reference mean. The LLM resembles History-only rather than ERP-inflated code, consistent with Section 6.4.

Run-to-run noise.. The within-pair cost range across the three original runs, divided by that pair's mean and averaged across pairs, is 10.5%; this is not a standard deviation. The runs disagree on beating History-only code for 34.1% of pairs. The LLM mean is cheaper than History-only for 17.1% of pairs and SAP-derived for 76.3%. Selecting each pair's best run with hindsight gives mean cost 2,765 (History-only: 2,764); selecting the worst gives 3,146. Such selection is not an implementable out-of-sample policy.

Plant C and segments.. Table 4 shows that the LLM advantage over SAP-derived parameters disappears in storage-bound pairs (+4.9% against −14.9% elsewhere) and reverses in Plant C (+33.3%); it does not vary monotonically with demand variability. Plant C is not an LLM-specific failure: the History-only code is also 28% more costly than SAP-derived parameters there, so the ERP parameters are leaner than any formula-based policy in that plant when lost sales are not charged. Plant C also has the least regular demand (median coefficient of variation of daily demand 0.61, lead-time coefficient of variation 0.78, against 0.51 and 0.63 over all pairs).

Table 4: Where the LLM differs from code. Original-scale mean cost per pair in the hard-limit replay with zero shortage penalty (LLM: mean of three original runs; History-only: textbook code; SAP: ERP-derived), relative costs (%), and the median smoothed reorder-point ratio $(R_{\text{LLM}} + 1)/(R_{\text{code}} + 1)$. Storage-bound pairs require downward projection because textbook safety stock plus order quantity exceeds capacity.

Segment	n	LLM	History-only	SAP	vs History-only (%)	vs SAP (%)	ROP ratio
All pairs	346	2,929	2,764	3,305	+6.0	-11.4	1.09
Plant A	215	2,370	2,231	2,782	+6.2	-14.8	1.08
Plant B	65	5,050	4,746	6,360	+6.4	-20.6	1.10
Plant C	66	2,662	2,549	1,997	+4.4	+33.3	1.11
Demand CV, lowest third	115	2,249	2,092	2,707	+7.5	-16.9	1.09
Demand CV, middle third	115	3,872	3,659	4,152	+5.8	-6.8	1.10
Demand CV, highest third	116	2,669	2,544	3,057	+4.9	-12.7	1.08
ERP safety stock zero	188	2,946	2,741	3,215	+7.5	-8.4	1.09
ERP safety stock positive	158	2,909	2,792	3,411	+4.2	-14.7	1.09
Storage-bound	145	1,491	1,408	1,422	+5.9	+4.9	1.10
Not storage-bound	201	3,967	3,743	4,663	+6.0	-14.9	1.08

6.8. Run-to-run stability

Original aggregate costs relative to SAP-derived are −11.4%, −11.3% and −11.4%. Figure 4 extends the ranking comparison to all six runs on shared masked-panel pairs. Individual policies still vary: original runs 2 and 3 have identical triples for only 36 of 365 pairs. Similar aggregate means do not establish item-level repeatability (Ouyang et al., 2025).

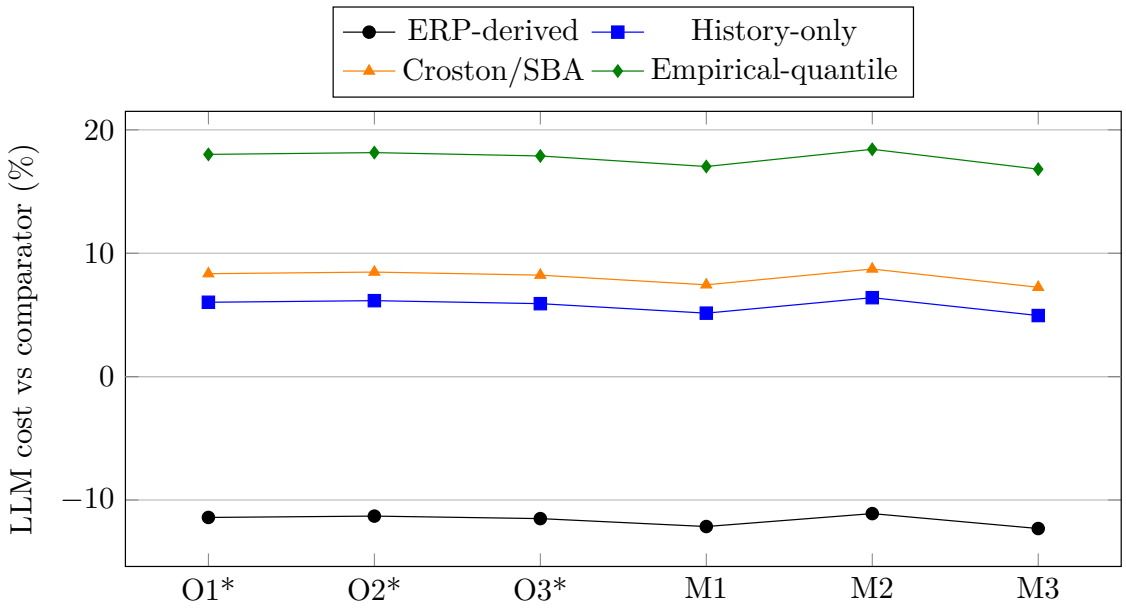


Figure 4: Ranking stability across six replays on 338 shared masked-panel pairs, zero shortage penalty. Negative values favour the LLM. O1*–O3* are rescaled original policies; M1–M3 are masked generations. Consistent signs indicate aggregate ranking stability; individual policies still vary.

6.9. Replication on the masked release

The raw data cannot be shared; an existing masked panel supports replay (see Data and Code Availability). Quantities are multiplied by secret per-pair factors and unit prices divided by those factors, preserving economic products before rounding. Identifiers are pseudonyms; dates, lead times and plant structure remain unchanged. This is pseudonymized, not demonstrated anonymous; unchanged dates and preserved economic products could still allow linkage to the source. Masking and employment establish neither redistribution rights nor permission. We repeated generation three times on this panel with the unchanged prompt (m1–m3). Run m1 produced 364 and m2/m3 all 365 policies (one pair exhausted retries in m1); 340, 342 and 342 were scorable, and 5, 4 and 4 invalid. Table 5 requires all seven comparators; SLT-informed and ERP-floor costs appear in Figure 2.

Table 5: Six-run comparison on 338 pairs feasible for all runs and seven comparators. Cost is mean holding-plus-ordering cost of masked-panel replay; fills and cost differences are percentages. O1*–O3* are rescaled frozen originals, M1–M3 fresh masked generations. Cohort inclusion, generation and rounding differ from Table 3; economic products survive reciprocal scaling.

Run	Cost	Mean fill	Aggr. fill	vs ERP	vs History	vs SBA	vs Empirical	vs Tuned
O1*	2,922	98.34	76.44	−11.4	+6.0	+8.3	+18.0	−10.6
O2*	2,925	98.11	76.27	−11.3	+6.2	+8.5	+18.2	−10.5
O3*	2,919	98.37	76.46	−11.5	+5.9	+8.2	+17.9	−10.7
M1	2,898	98.33	76.34	−12.2	+5.1	+7.4	+17.0	−11.3
M2	2,932	98.37	76.38	−11.1	+6.4	+8.7	+18.4	−10.3
M3	2,892	98.37	76.29	−12.3	+4.9	+7.2	+16.8	−11.5

On each masked run's own feasible cohort, the LLM is 11–12% cheaper than ERP-derived parameters and 5–6%, 7–9% and 16–18% costlier than History-only, Croston/SBA and empirical-quantile code. Matched costs relative to rescaled original run 1 are 0.991, 1.004 and 0.991, not comparisons with confidential unscaled means. Table 5 and Figures 2–4 use the same 338 pairs; Figure 3 confirms the PL3 reversal. Quantity scaling changes volume-fill weights; generation, cohort inclusion and integer rounding can change cost summaries despite preserved economic products. Repeated processing does not validate source grain or establish monetary confidentiality.

6.10. Sensitivity analyses

Plant C calendar.. Plant C's working-day flag required reconstruction, so we re-scored all policies under four alternatives (any-plant, strict-all, exclusion of Plant C, calendar-day arrivals). Where Plant C can be scored, the LLM remains 31–35% more expensive than SAP-derived there, so the Plant C result is not an artefact of the repair.

Shortage penalty.. Rescoring the original fixed trajectories at one and five times unit cost makes the simulation-tuned policy about 33% cheaper than SAP-derived; LLM run 1 costs 2.4% more than SAP-derived at both multiples. These are accounting sensitivities, not policy optimization at each penalty. Table 6 retains the original-scale analysis; the public six-run evidence also records shortage exposure for masked-scale rescoring.

Break-even shortage price.. Table 6 re-scores the same replays for a lost-sales penalty between zero and one unit cost per unit. The LLM is cheaper than SAP-derived parameters only while the penalty stays below 0.9% of the unit cost, and cheaper than the History-only and empirical-quantile code only above 1.5% and 2.4%, by about 1% or less. It is cheaper than the SLT-informed and ERP-floor ORs at every penalty and cheaper than the simulation-tuned policy only below 0.06%. From a penalty of 1% of the unit cost, the SAP-derived, History-only, Croston/SBA, empirical-quantile and LLM policies lie within 4% of each other, while the simulation-tuned policy is 28–33% cheaper than SAP-derived parameters. The ranking between the LLM and the formulas is thus decided by a shortage price that the primary basis sets to zero, which is why it is reported as a sensitivity and not as a headline result.

Secondary strict-admission validation.. Supplementary File S2 separately replays the same 338 pairs after reproducing frozen clipping results. Fixed policies use floored physical headroom and defer releases below MOQ. Strict replay has no sub-MOQ releases or capacity

Table 6: Break-even shortage price. Mean total cost per pair (holding, ordering and shortage cost) in the hard-limit replay when each lost unit is charged at the given multiple of the unit cost, and the break-even multiple at which the LLM and the arm have equal mean cost (common-feasible cohort, $n = 346$).

Arm	0	0.01	0.1	1	Break-even multiple of unit cost
SAP-derived	3,305	19,613	166,388	1,634,136	0.0089; LLM cheaper below
History-only OR	2,764	19,609	171,207	1,687,191	0.0145; LLM cheaper above
Croston/SBA OR	2,705	19,427	169,927	1,674,925	none; LLM costlier on the grid
Empirical-quantile OR	2,489	19,402	171,618	1,693,784	0.0242; LLM cheaper above
Simulation-tuned	3,266	14,174	112,344	1,094,043	0.0006; LLM cheaper below
LLM (mean of three runs)	2,929	19,660	170,238	1,676,014	–

breaches; M1 mean fill falls from 98.326% to 98.323%, with 48 deferrals, and Plant C cost remains above ERP-derived (2,599 versus 1,845 masked units). A conditional admission invariant and independent finite-state/counterexample tests support the simulator property, not service or optimality. Initialization is hypothetical; opening excess is retained and excluded from the invariant, not descriptive summaries (none on this cohort). Both modes report cost, mean/volume fill and adverse outcomes. Separately retuned arms do not match held-out service.

7. Discussion, Limitations and Research Perspectives

What the evidence shows.. The empirical finding connects three limits of ERP-relative savings: stronger classical comparators are cheaper at zero shortage price, Plant C and fixed-trajectory shortage repricing reverse the pooled comparison, and static-valid proposals fail joint trajectory constraints. Passing one comparison does not resolve the others. All original proposals passed the static gate; masked generations include missing and invalid outputs. S2 isolates conditional admission support, not competitive performance or equal-service superiority. These linked observations define the screening a proposal must pass before planner review.

Practitioner feedback.. Planners at the case company formally reviewed the proposals and compared them in several internal benchmarks; their feedback was positive. They valued receiving explainable, per-material reorder points and lot sizes directly from production data,

without model training or new infrastructure. They regarded the gate, trajectory replay and escalation rules as a practical path from LLM proposals to planning processes. This feedback is qualitative and author-reported; it is not part of the quantitative evidence above.

Implications for practice.. (1) Benchmark against data-driven code, not ERP alone; input treatment matters. (2) Price shortages before interpreting cost rankings (Section 6.10). (3) Verify storage and MOQ jointly: static compliance and capacity clipping do not ensure both. (4) Expect plant differences; define review/escalation rules (Parasuraman & Manzey, 2010). (5) Version prompts, retain provenance and repeat generations. (6) No project-specific training or fine-tuning was needed. Session logs give a median of about 5–6 minutes per material; from prompt and output lengths at list prices, inference costs an estimated few cents (about US$0.03–0.10) per material and generation, not billed fees.

Limitations.. The evidence comes from one manufacturer's three-plant panel. Model identity is operator-declared; planner feedback is qualitative and cost is estimated; fixed-policy comparisons are not equal-service; separately retuned arms do not match held-out service. Policies are single-item (r, Q), and trained neural or reinforcement-learning baselines are left to future work.

Research perspectives.. (1) *Constraint-aware hybrids:* test mechanism-aware generation with independent joint-feasibility checks, motivated by (Yan & Ji, 2026) and (Dehghani, 2026), without transferring performance claims. Policy aggregation is untested here, distinct from spare-parts forecasting (Abbaszadeh Nakhost et al., 2025); LLM review of data-driven policies (Wasserkrug et al., 2025) also needs evaluation. (2) *Matched comparisons:* compare InvEvolve, trained inventory-control frameworks (Gabellini et al., 2026), reinforcement learning (Kolyaei et al., 2025; Stranieri et al., 2024) and richer (s, S) policies (Babai et al., 2026; Prak

et al., 2021) at matched information, service and measured setup/inference/review budgets. (3) *Planner trials:* measure time, overrides, realized service/inventory and adoption (Cannas et al., 2024; Trifone et al., 2026; Venkatesh et al., 2024) prospectively; published guarantees elsewhere do not transfer to InventOR.

8. Conclusion

InventOR shows that an unmodified pretrained LLM, without fine-tuning or training data, produces industrial replenishment proposals worth screening, and which checks decide when its ERP-relative savings justify operational consideration. The original 346-pair comparison gives about 11% lower cost at zero shortage price and slightly lower mean service, but classical baselines are cheaper and Plant C reverses the pooled ERP ranking. Repricing fixed trajectories removes the ERP-relative saving above approximately 0.9% of unit cost. Static checks and capacity clipping miss joint trajectory constraints; S2 supplies conditional admission support, not a service guarantee. A public masked panel supports numerical replay at an estimated few cents per material; quantitative planner trials and further industrial sites are the next steps.

Appendix A. Formal Definitions of the Reference Calculation and Simulation

Reference calculation.. These are standard textbook calculations (Axsäter, 2015; Silver et al., 2017). For pair i, $\bar{d}_i, \sigma_{D,i}$ describe pre-cutoff working-day demand; $L_i, \sigma_{L,i}$ describe recorded lead times; K_i is ordering cost, h_i annual holding cost per unit, and z_i the normal service-target quantile. Set $Q_i = \max\{\mathrm{MOQ}_i, \lceil\sqrt{2K_i \cdot 250\bar{d}_i/h_i}\,\rceil\}$, $\sigma_{LTD,i} = \sqrt{L_i^{\mathrm{eff}}\sigma_{D,i}^2 + \bar{d}_i^2\sigma_{L,i}^2}$, $SS_i = \lceil z_i\sigma_{LTD,i}\rceil$ and $R_i = \lceil \bar{d}_i L_i^{\mathrm{eff}} + SS_i\rceil$. Here $L_i^{\mathrm{eff}} = L_i$ for History-only and $L_i + \max(\mathrm{ERP\ SLT}_i, 0)$ for SLT-informed and ERP-floor; the latter also floors SS_i at ERP safety stock. SAP-derived uses ERP safety stock, the same Q_i, and ERP safety lead time in its reorder point. Projection enforces $SS_i + Q_i \le \texttt{max_storage_units}_i$ when MOQ plus safety stock fits.

Simulation.. Review is discrete, not event-driven continuous review. Each post-cutoff working observation receives due orders, serves demand, loses unmet demand, then orders at most

once if position (on hand plus on order) $\leq R_i$. Rounded mean lead time is at least one subsequent working observation. Historical receipts/corrections/scrap are ignored. Cost is $TC_i = h_i \bar{I}_i T/250 + K_i N_i$: $\bar{I}_i$ averages end-observation stock over T working observations; N_i counts placed orders, including arrivals beyond the horizon; no terminal salvage is applied. Fill is served/recorded units (100% for zero recorded demand); stockout days count unmet-demand working observations. Positive hard-limit orders are $\min\{Q_i, \text{capacity}_i - \lfloor \text{position}_i \rfloor\}$, even below MOQ; otherwise none. Observed opening stock is not clipped.

Appendix B. System Prompt Excerpts

The complete prompts are verbatim in Supplementary File S1. Key instructions: "You are an autonomous inventory analyst" who analyses one material with code and saves a JSON policy; forbidden are "No Monte Carlo / np.random," distribution fitting, ML libraries and "No Croston, ADI, Syntetos-Boylan"; `max_storage_units`> 0 is a hard limit, safety_stock+ order_quantity $\leq$ `max_storage_units`, and a MOQ-capacity conflict must be escalated, not silently emitted; policy values are conservative integers (`math.ceil`).

Author Contributions

Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Validation, Visualization, Writing – original draft, Writing – review and editing: Aris Dressino.

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Declaration of Competing Interest

The author reports an employment relationship with an industrial manufacturer (name withheld). The author declares no other competing interests.

Data and Code Availability

Raw operational data are confidential. A masked panel contains the same 233,078 records, with pseudonymous identifiers and secret per-pair quantity and price factors. The panel, rescaled frozen policies, three masked generations, sanitized provenance, scoring code, aggregate evidence and manuscript sources are at `github.com/DressPD/inventor_public`. The archive version DOI is `https://doi.org/10.5281/zenodo.23627916` (software v1.3.0). That archive preserves the evaluated evidence; subsequent manuscript reporting corrections do not amend its frozen source snapshot. Supplementary File S1 contains the prompt (excerpts in Appendix B). Runner-hash amendments are recorded separately from original call provenance, with retained original hashes and amendment histories; they must not be read as

original execution hashes. Model identity (Claude Sonnet 4.6) is operator-declared; runtime identifier and sampling settings are unavailable.

Supplementary File S2 reports the strict-admission methods, invariant, tests and results; its code, masked inputs and frozen policies are in the public repository above. This later support is distinct from the unchanged archives; it supports replay, not proprietary inference reconstruction.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

OpenCode AI-assisted tools, including Claude Sonnet 5.5 and GPT-6.1 Sol, assisted language editing, restructuring, literature search and reference selection, consistency and citation checks, LaTeX preparation and analysis-script development. These preparation tools are distinct from the operator-declared Claude Sonnet 4.6 deployment evaluated in the study. The author retains responsibility for reviewing and verifying the resulting text and code and for the content of the final article. Figures are deterministic diagrams and data plots rendered from source files, not image-model outputs.

References

Abbaszadeh Nakhost, M., Bloemer, A., & Minner, S. (2025). Full automation or just a copilot? a study of large language models on spare parts demand forecasting with OpenAI o1-mini [Advance online publication]. *International Journal of Production Research*. https://doi.org/10.1080/00207543.2025.2591225

Abolghasemi, M., Ganbold, O., & Rotaru, K. (2025). Humans vs. large language models: Judgmental forecasting in an era of advanced AI. *International Journal of Forecasting*, *41*(2), 631–648. https://doi.org/10.1016/j.ijforecast.2024.07.003

Ahmaditeshnizi, A., Gao, W., & Udell, M. (2024). OptiMUS: Scalable optimization modeling with (MI)LP solvers and large language models. *Proceedings of the 41st International Conference on Machine Learning*, *235*, 577–596. https://proceedings.mlr.press/v235/ahmaditeshnizi24a.html

Amarasinghe, P., Nguyen, S., Sun, Y., Arisian, S., & Alahakoon, D. (2026). Business optimization for digital manufacturing: A fine-tuned large language model approach. *International Journal of Production Economics*, *295*, 109934. https://doi.org/10.1016/j.ijpe.2026.109934

Amershi, S., Weld, D., Vorvoreanu, M., Fourney, A., Nushi, B., Collisson, P., Suh, J., Iqbal, S., Bennett, P. N., Inkpen, K., Teevan, J., Kikin-Gil, R., & Horvitz, E. (2019). Guidelines for human-AI interaction. *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, 1–13. https://doi.org/10.1145/3290605.3300233

Axsäter, S. (2015). *Inventory control* (3rd ed., Vol. 225). Springer. https://doi.org/10.1007/978-3-319-15729-0

Babai, M. Z., Syntetos, A. A., & Teunter, R. H. (2026). Fifty years of inventory research from a forecasting perspective. *European Journal of Operational Research*, *331*(1), 1–20. https://doi.org/10.1016/j.ejor.2025.07.003

Baek, J., Fu, Y., Ma, W., & Peng, T. (2026). AI agents for inventory control: Human-LLM-OR complementarity. *arXiv preprint arXiv:2602.12631*. https://doi.org/10.48550/arXiv.2602.12631

Bai, L., Alexopoulos, C., Ferguson, M. E., & Tsui, K.-L. (2012). A simple and robust batch-ordering inventory policy under incomplete demand knowledge. *Computers & Industrial Engineering*, *63*, 343–353. https://doi.org/10.1016/j.cie.2012.02.010

Bijvank, M., & Vis, I. F. A. (2011). Lost-sales inventory theory: A review. *European Journal of Operational Research*, *215*(1), 1–13. https://doi.org/10.1016/j.ejor.2011.02.004

Cannas, V. G., Ciano, M. P., Saltalamacchia, M., & Secchi, R. (2024). Artificial intelligence in supply chain and operations management: A multiple case study research. *International Journal of Production Research*, *62*(9), 3333–3360. https://doi.org/10.1080/00207543.2023.2232050

Croston, J. D. (1972). Forecasting and stock control for intermittent demands. *Operational Research Quarterly*, *23*(3), 289–303. https://doi.org/10.1057/jors.1972.50

Dehghani, M. (2026). SimGPT: Unlocking potentials of generative AI in simulation via Simio and LLM integration. *Computers & Industrial Engineering*, *218*, 112104. https://doi.org/10.1016/j.cie.2026.112104

DeHoratius, N., & Raman, A. (2008). Inventory record inaccuracy: An empirical analysis. *Management Science*, *54*(4), 627–641. https://doi.org/10.1287/mnsc.1070.0789

Dietvorst, B. J., Simmons, J. P., & Massey, C. (2015). Algorithm aversion: People erroneously avoid algorithms after seeing them err. *Journal of Experimental Psychology: General*, *144*(1), 114–126. https://doi.org/10.1037/xge0000033

El Baz, H., Ghali, M.-K., Andrus, L., Won, D., Yoon, S. W., & Wang, Y. (2026). Generative AI-based optimization for high-mix warehouses: Integrating agentic LLMs and discrete event simulation. *Computers & Industrial Engineering*, *214*, 111884. https://doi.org/10.1016/j.cie.2026.111884

Fosso Wamba, S., Guthrie, C., Queiroz, M. M., & Minner, S. (2024). ChatGPT and generative artificial intelligence: An exploratory study of key benefits and challenges in operations and supply chain management. *International Journal of Production Research*, *62*(16), 5676–5696. https://doi.org/10.1080/00207543.2023.2294116

Fu, W., & Chien, C.-F. (2019). UNISON data-driven intermittent demand forecast framework to empower supply chain resilience and an empirical study in electronics distribution. *Computers & Industrial Engineering*, *135*, 940–949. https://doi.org/10.1016/j.cie.2019.07.002

Gabellini, M., Regattieri, A., Bortolini, M., & Ronchi, M. (2026). A novel data-driven approach for integrated inventory replenishment, safety stock and lead time optimisation [Advance online publication]. *International Journal of Production Research*. https://doi.org/10.1080/00207543.2026.2696440

Gorry, G. A., & Scott Morton, M. S. (1971). A framework for management information systems. *Sloan Management Review*, *13*(1), 55–70.

Grasso, E. T., & Taylor, B. W. (1984). A simulation-based experimental investigation of supply/timing uncertainty in MRP systems. *International Journal of Production Research*, *22*(3), 485–497. https://doi.org/10.1080/00207548408942468

Guo, W., Ren, Z., Du, W., & Weng, T. (2025). TFT-MPIR: An end-to-end multi-period inventory replenishment strategy based on temporal fusion transformer. *Expert Systems with Applications*, *261*, 125464. https://doi.org/10.1016/j.eswa.2024.125464

Hadley, G., & Whitin, T. M. (1963). *Analysis of inventory systems*. Prentice-Hall.

Heese, H. S. (2025). Minimum order quantity requirements in the presence of supply disruption risk and retail competition. *International Journal of Production Research*, *63*(16), 6056–6070. https://doi.org/10.1080/00207543.2025.2468391

Hewamalage, H., Ackermann, K., & Bergmeir, C. (2023). Forecast evaluation for data scientists: Common pitfalls and best practices. *Data Mining and Knowledge Discovery*, *37*(2), 788–832. https://doi.org/10.1007/s10618-022-00894-5

Huang, C., Lin, J., Tang, Z., Jiang, B., Jiang, R., Wang, B., & Wei, L. (2026). *InvEvolve: Evolving white-box inventory policies via large language models with performance guarantees* [Preprint, version 4, 5 June 2026]. arXiv: 2605.00369v4. https://arxiv.org/abs/2605.00369v4

Huang, C., Tang, Z., Hu, S., Jiang, R., Zheng, X., Ge, D., Wang, B., & Wang, Z. (2025). ORLM: A customizable framework in training large models for automated optimization modeling. *Operations Research*, *73*(6), 2986–3009. https://doi.org/10.1287/opre.2024.1233

Jackson, I., Ivanov, D., Dolgui, A., & Namdar, J. (2024). Generative artificial intelligence in supply chain and operations management: A capability-based framework for analysis and implementation. *International Journal of Production Research*, *62*(17), 6120–6145. https://doi.org/10.1080/00207543.2024.2309309

Jannelli, V., Schöpf, S., Bickel, M., Netland, T., & Brintrup, A. (2025). Agentic LLMs in the supply chain: Towards autonomous multi-agent consensus-seeking [Advance online publication]. *International Journal of Production Research*. https://doi.org/10.1080/00207543.2025.2604311

Ji, Z., Lee, N., Frieske, R., Yu, T., Su, D., Xu, Y., Ishii, E., Bang, Y. J., Madotto, A., & Fung, P. (2023). Survey of hallucination in natural language generation. *ACM Computing Surveys*, *55*(12), 1–38. https://doi.org/10.1145/3571730

Kang, Y., & Gershwin, S. B. (2005). Information inaccuracy in inventory systems: Stock loss and stockout. *IIE transactions*, *37*(9), 843–859. https://doi.org/10.1080/07408170590969861

Kolyaei, M., Zhang, L., & Blom, M. (2025). Inventory replenishment and fulfilment decisions for an omnichannel retailer: A reinforcement learning-based method. *International Journal of Production Research*, *63*(24), 9571–9592. https://doi.org/10.1080/00207543.2025.2520596

Kourentzes, N. (2013). Intermittent demand forecasts with neural networks. *International Journal of Production Economics*, *143*(1), 198–206. https://doi.org/10.1016/j.ijpe.2013.01.009

Kourentzes, N., Trapero, J. R., & Barrow, D. K. (2020). Optimising forecasting models for inventory planning. *International Journal of Production Economics*, *225*, 107597. https://doi.org/10.1016/j.ijpe.2019.107597

Lau, R. S. M., Xie, J., & Zhao, X. (2008). Effects of inventory policy on supply chain performance: A simulation study of critical decision parameters. *Computers & Industrial Engineering*, *55*, 620–633. https://doi.org/10.1016/j.cie.2008.02.002

Li, B., Mellou, K., Zhang, B., Pathuri, J., & Menache, I. (2023). Large language models for supply chain optimization. *arXiv preprint arXiv:2307.03875*. https://doi.org/10.48550/arXiv.2307.03875

Li, L., Liu, Y., Jin, Y., Cheng, T. C. E., & Zhang, Q. (2024). Generative AI-enabled supply chain management: The critical role of coordination and dynamism. *International Journal of Production Economics*, *277*, 109388. https://doi.org/10.1016/j.ijpe.2024.109388

Liu, M., Bai, Y., Qi, M., & Shen, Z.-J. M. (2026). Inventory management with transformer: Automated decision making for order timing and quantity. *Service Science*, *18*(3), 196–211. https://doi.org/10.1287/serv.2024.0236

Mersereau, A. J. (2013). Information-sensitive replenishment when inventory records are inaccurate. *Production and Operations Management*, *22*(4), 792–810. https://doi.org/10.1111/j.1937-5956.2011.01305.x

Ouyang, S., Zhang, J. M., Harman, M., & Wang, M. (2025). An empirical study of the non-determinism of ChatGPT in code generation. *ACM Transactions on Software Engineering and Methodology*, *34*(2), 1–28. https://doi.org/10.1145/3697010

Parasuraman, R., & Manzey, D. H. (2010). Complacency and bias in human use of automation: An attentional integration. *Human Factors*, *52*(3), 381–410. https://doi.org/10.1177/0018720810376055

Parasuraman, R., Sheridan, T. B., & Wickens, C. D. (2000). A model for types and levels of human interaction with automation. *IEEE Transactions on Systems, Man, and Cybernetics – Part A: Systems and Humans*, *30*(3), 286–297. https://doi.org/10.1109/3468.844354

Porteus, E. L. (2002). *Foundations of stochastic inventory theory*. Stanford University Press. https://doi.org/10.1515/9781503619883

Prak, D., Teunter, R., Babai, M. Z., Boylan, J. E., & Syntetos, A. (2021). Robust compound Poisson parameter estimation for inventory control. *Omega*, *104*, 102481. https://doi.org/10.1016/j.omega.2021.102481

Quan, Y., & Liu, Z. (2024). InvAgent: A large language model based multi-agent system for inventory management in supply chains. *arXiv preprint arXiv:2407.11384*. https://doi.org/10.48550/arXiv.2407.11384

Raji, I. D., Smart, A., White, R. N., Mitchell, M., Gebru, T., Hutchinson, B., Smith-Loud, J., Theron, D., & Barnes, P. (2020). Closing the AI accountability gap: Defining an end-to-end framework for internal algorithmic auditing. *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*, 33–44. https://doi.org/10.1145/3351095.3372873

Ren, K., Bidkhori, H., & Shen, Z.-J. M. (2024). Data-driven inventory policy: Learning from sequentially observed non-stationary data. *Omega*, *123*, 102942. https://doi.org/10.1016/j.omega.2023.102942

Rostami-Tabar, B., & Disney, S. M. (2023). On the order-up-to policy with intermittent integer demand and logically consistent forecasts. *International Journal of Production Economics*, *257*, 108763. https://doi.org/10.1016/j.ijpe.2022.108763

Sahoo, P., Singh, A. K., Saha, S., Jain, V., Mondal, S., & Chadha, A. (2024). A systematic survey of prompt engineering in large language models: Techniques and applications. *arXiv preprint arXiv:2402.07927*. https://doi.org/10.48550/arXiv.2402.07927

Schett, G., Ehrig, C., Schwendinger, B., Neumann, U., Beck, N., Stadler, F., & Ansari, F. (2026). Integrated demand forecasting and reinforcement learning for order point optimization in inventory planning. *International Journal of Production Economics*, *299*, 110111. https://doi.org/10.1016/j.ijpe.2026.110111

Shim, J. P., Warkentin, M., Courtney, J. F., Power, D. J., Sharda, R., & Carlsson, C. (2002). Past, present, and future of decision support technology. *Decision Support Systems*, *33*(2), 111–126. https://doi.org/10.1016/S0167-9236(01)00139-7

Silver, E. A., Pyke, D. F., & Thomas, D. J. (2017). *Inventory and production management in supply chains* (4th ed.). CRC Press. https://doi.org/10.1201/9781315374406

Song, Z., Xie, Y., Yang, L., & Zhao, Y. (2026). Large language models in supply chain management: A systematic literature review and application framework. *International Journal of Production Research*, *64*(16), 7144–7184. https://doi.org/10.1080/00207543.2026.2641103

Sprague, R. H. (1980). A framework for the development of decision support systems. *MIS Quarterly*, *4*(4), 1–26. https://doi.org/10.2307/248957

Stranieri, F., Stella, F., & Kouki, C. (2024). Performance of deep reinforcement learning algorithms in two-echelon inventory control systems. *International Journal of Production Research*, *62*(17), 6211–6226. https://doi.org/10.1080/00207543.2024.2311180

Svetunkov, I., & Boylan, J. E. (2023). iETS: State space model for intermittent demand forecasting. *International Journal of Production Economics*, *265*, 109013. https://doi.org/10.1016/j.ijpe.2023.109013

Syntetos, A. A., & Boylan, J. E. (2001). On the bias of intermittent demand estimates. *International Journal of Production Economics*, *71*(1–3), 457–466. https://doi.org/10.1016/S0925-5273(00)00143-2

Syntetos, A. A., & Boylan, J. E. (2006). On the stock control performance of intermittent demand estimators. *International Journal of Production Economics*, *103*(1), 36–47. https://doi.org/10.1016/j.ijpe.2005.04.004

Syntetos, A. A., & Boylan, J. E. (2005). The accuracy of intermittent demand estimates. *International Journal of Forecasting*, *21*(2), 303–314. https://doi.org/10.1016/j.ijforecast.2004.10.001

Tashman, L. J. (2000). Out-of-sample tests of forecasting accuracy: An analysis and review. *International Journal of Forecasting*, *16*(4), 437–450. https://doi.org/10.1016/S0169-2070(00)00065-0

Thürer, M., Fernandes, N. O., Haeussler, S., & Stevenson, M. (2023). Dynamic planned lead times in production planning and control systems: Does the lead time syndrome matter? *International Journal of Production Research*, *61*(4), 1268–1282. https://doi.org/10.1080/00207543.2022.2034193

Trifone, N., Dallari, F., & Brintrup, A. (2026). Drivers and constraints of agentic AI in supply chain management: An exploratory study [Advance online publication]. *International Journal of Production Research*. https://doi.org/10.1080/00207543.2026.2669878

van Kampen, T. J., van Donk, D. P., & van der Zee, D.-J. (2010). Safety stock or safety lead time: Coping with unreliability in demand and supply. *International Journal of Production Research*, *48*(24), 7463–7481. https://doi.org/10.1080/00207540903348346

Venkatesh, V., Raman, R., & Cruz-Jesus, F. (2024). AI and emerging technology adoption: A research agenda for operations management. *International Journal of Production Research*, *62*(15), 5367–5377. https://doi.org/10.1080/00207543.2023.2192309

Wang, Q., Chen, Y., Mi, Y., She, C., Zhao, P., & Chen, C. (2026). Contextual distributionally robust optimisation for inventory replenishment optimisation under uncertain demand. *International Journal of Production Research*, *64*(19), 8393–8411. https://doi.org/10.1080/00207543.2026.2653131

Wang, X., Rodrigues, V. S., Demir, E., & Sarkis, J. (2024). Algorithm aversion during disruptions: The case of safety stock. *International Journal of Production Economics*, *278*, 109442. https://doi.org/10.1016/j.ijpe.2024.109442

Wang, Y., Wang, J., & Chu, Z. (2025). Multi-agent large language models as evolutionary optimizers for scheduling optimization. *Computers & Industrial Engineering*, *206*, 111197. https://doi.org/10.1016/j.cie.2025.111197

Wasserkrug, S., Boussioux, L., Den Hertog, D., Mirzazadeh, F., Birbil, Ş. İ., Kurtz, J., & Maragno, D. (2025). Enhancing decision making through the integration of large language models and operations research

optimization. *Proceedings of the AAAI Conference on Artificial Intelligence*, *39*(27), 28643–28650. https://doi.org/10.1609/aaai.v39i27.35090

Watts, S., Shankaranarayanan, G., & Even, A. (2009). Data quality assessment in context: A cognitive perspective. *Decision Support Systems*, *48*(1), 202–211. https://doi.org/10.1016/j.dss.2009.07.012

Wu, R. (2026). Inventory optimization under supply chain disruptions: Leveraging large language models for human-AI collaborative decision-making. *Journal of King Saud University Computer and Information Sciences*, *38*(2), 73. https://doi.org/10.1007/s44443-025-00458-9

Yan, J., & Ji, S. (2026). Industrial mechanism-aware multi-agent collaboration of large language models for job shop scheduling optimization. *Computers & Industrial Engineering*, *217*, 112079. https://doi.org/10.1016/j.cie.2026.112079

Zabraoui, O., Hmamou, Y., Chafi, A., & Kammouri Alami, S. (2026). Conversational AI for inventory analytics: Leveraging large language models in supply chain management. *Computers & Industrial Engineering*, *220*, 112298. https://doi.org/10.1016/j.cie.2026.112298

Zhang, B., Luo, P., Yang, G., Soong, B.-H., & Yuen, C. (2025). OR-LLM-Agent: Automating modeling and solving of operations research optimization problems with reasoning LLM. *arXiv preprint arXiv:2503.10009*. https://doi.org/10.48550/arXiv.2503.10009

Zhang, X., Dai, W., & Cai, X. (2024). Inventory replenishment decisions with uncertain price and demand. *International Journal of Production Research*, *62*(7), 2663–2682. https://doi.org/10.1080/00207543.2023.2223701

Zipkin, P. H. (2000). *Foundations of inventory management*. McGraw-Hill.